

Lecture notes: Chapter 8 (Strogatz)

Week of 09/29: no lectures

Bifurcations in 2-Dimensions

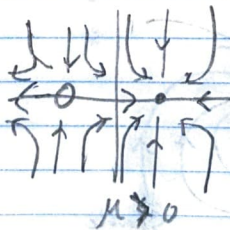
→ fixed points / closed orbits can be created or destroyed by varying parameters

→ Bifurcations occur when the topological structure of the phase portrait changes due to a parameter being varied

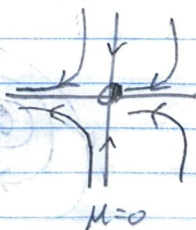
→ Higher-D bifurcations are qualitatively analogous to the 1-D case: the critical point just becomes a one-dimensional subspace along which the bifurcations occur

• Saddle-node Bifurcation

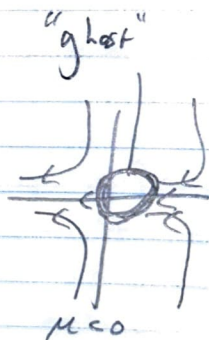
$$\begin{cases} \dot{x} = \mu - x^2 \\ \dot{y} = -y \end{cases}$$



$\mu > 0$



$\mu = 0$



$\mu < 0$

$$x^*, y^* = (\pm\sqrt{\mu}, 0)$$

$$x^*, y^* = (0, 0)$$

none

"ghost"

• Transcritical and Pitchfork Bifurcations

$$\dot{x} = \mu x - x^2$$

$$\dot{y} = -y$$

transcritical

$$\dot{x} = \mu x - x^3$$

$$\dot{y} = -y$$

supercritical p-fork

$$\dot{x} = \mu x + x^3$$

$$\dot{y} = -y$$

subcritical p-fork

• Hopf Bifurcation

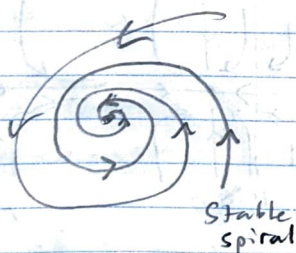
→ Consider the scenario where 2 complex conjugate eigenvalues simultaneously cross the imaginary axis into the right half-plane

→ This physically arises in a system that settles to equilibrium under damped oscillations:

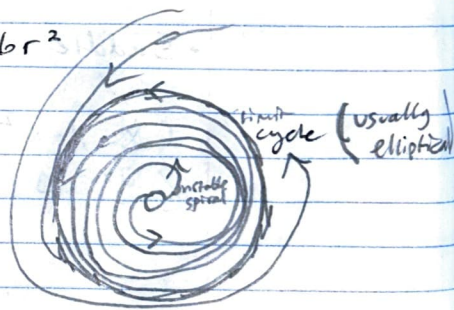
- Small disturbances decay
- Let decay rate depend on parameter μ
- let decay transition to growth @ $\mu = \mu_c$
- Oscillations grow unboundedly, unstable system
- yields small-scale, sinusoidal limit cycle oscillations

→ For example:

$$\dot{r} = \mu r - r^3, \quad \dot{\theta} = \omega + br^2$$



$$\mu < 0$$



$$\mu > 0$$

→ generic properties of Supercritical Hopf bifurcation:

- the size of the limit cycle grows continuously from zero, increases $\propto \sqrt{\mu - \mu_c}$ for $|\mu - \mu_c| \ll 1$
 - The approximate frequency of the limit cycle is $\omega = \text{Im} \lambda$, evaluated at $\mu = \mu_c$.
- $\Rightarrow T = 2\pi / \text{Im} \lambda + \mathcal{O}(\mu - \mu_c)$

→ subcritical case:

- more dramatic: after bifurcation, trajectories jump to a distant attractor (Lorenz attractor system, e.g.)
- trajectories may diverge to infinity

$$\dot{r} = \mu r + r^3 - r^5, \quad \dot{\theta} = \omega + br^2$$

→ degenerate case

- fixed point @ origin can change stability, but no limit cycle develops

- Global bifurcations also exist, where cycles (rather than FP's) are created/destroyed